

University of Natural Resources and Life Sciences, Vienna (BOKU)

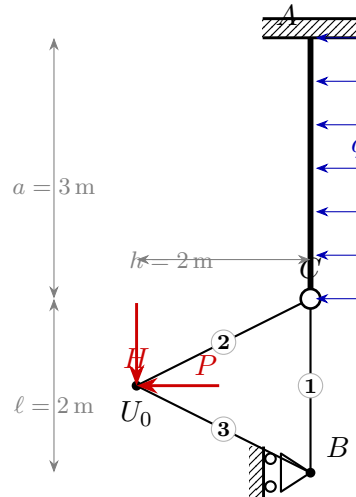
VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

Date: 26.06.2026

Given system

The system consists of two rigid bodies connected by a hinge at C : a bending beam $A-C$ (fixed support at A) and a truss (roller support at B).



Given quantities: $a = 3 \text{ m}$ $\ell = 2 \text{ m}$ $h = 2 \text{ m}$ $q = 5 \text{ kN/m}$ $P = 10 \text{ kN}$ $H = 8 \text{ kN}$

Tasks

1. Determine the degree of static determinacy of the system.
2. Compute all support reactions and the hinge force at C .
3. Determine all member forces of the truss (state tension/compression).
4. Draw the section-force diagrams $N(x)$, $V(x)$ and $M(x)$ for the bending beam $A-C$.